SDLC Copilot — Project Overview

> A plain-language guide for anyone who wants to understand what SDLC Copilot is, why it exists, and how it works — no coding knowledge required.

What is SDLC Copilot?

SDLC Copilot is an **AI-powered assistant for software development teams**. It connects to your GitHub repositories and uses the power of large language models (like GPT or Claude) to help teams understand their code, answer project questions, and write better requirement documents — all in real time through a web browser.

Think of it as having a knowledgeable team member available 24/7 who has read all your project documentation and can answer questions, explain how things work, and help your analysts write clear requirement documents.

The Four Things It Does

1. File Summarizer

"What does this file do?"

Give the assistant a GitHub repository address and a file path. Within seconds, it reads the file and explains it in plain English — whether the file is a complex piece of code, a configuration file, or a long markdown document.

```
You type:   facebook / react / packages/react/README.md
You get:    A structured summary explaining what the file contains,
            its purpose, key sections, and important notes.
```

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2. Knowledge Base Chat

"How does [feature/module/process] work?"

Connect the assistant to your project's documentation repository (a GitHub repo containing your markdown files, guides, and notes). Once connected, you can ask it any question in plain English and it will:

- Search through the relevant documentation files
- Answer based only on your actual project docs (not guessing)
- Ask a clarifying question if your question is ambiguous
- Tell you honestly if the information isn't in the docs

```
You ask:    "How does the user authentication flow work?"
It answers: Reads the INDEX file, fetches auth-module.md from your
            knowledge base, and gives a detailed, cited answer.
```

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3. UD Creator (Understanding Document Creator)

"Help me write a requirement document."

This is the most powerful standalone feature. A "UD" (Understanding Document) is a structured document that captures the full details of a software requirement before developers start building it. Creating a good UD normally takes a lot of back-and-forth between analysts and developers.

The UD Creator automates this process:

```
Step 1:  You describe a requirement (paste a ticket, write a few sentences)
Step 2:  The AI asks you 3–7 targeted clarifying questions
Step 3:  You answer; it may ask one more focused round if needed
Step 4:  It produces a complete UD draft, marking anything uncertain with ⚠
Step 5:  You review and give feedback ("Change section 3", "Looks good")
Step 6:  It revises and shows what changed
Step 7:  You say "approved" — it generates a downloadable Markdown file
```

The AI uses your project's knowledge base during this process, so the UD it produces is tailored to your actual systems, naming conventions, and existing components.

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4. SDLC Agent (Azure DevOps Automation)

"Automate the full SDLC lifecycle from ticket to attached UD."

The SDLC Agent combines work item management with UD creation into a single automated workflow. It connects to your Azure DevOps project and handles the entire cycle without copy-pasting between tools:

```
Step 1:  You enter an Azure DevOps Work Item ID (e.g. 1234)
Step 2:  The AI fetches the work item — title, description, acceptance criteria,
state
Step 3:  It shows a summary card and asks you to confirm before proceeding
Step 4:  It asks 3–7 targeted questions (supplementing what’s already in the
ticket)
Step 5:  It drafts a complete UD, auto-populating fields from the work item
Step 6:  You review and give feedback
Step 7:  You say "approved" — it generates a downloadable UD file AND
Step 8:  Uploads the UD file directly to Azure DevOps as an attachment on the work
item
```

The result is a fully traceable link: the work item in Azure DevOps has the UD attached, the analyst has a local copy, and the entire process happened through a single chat session.

How It All Works — The Big Picture

```
┌──────────────────────────────────────┐
│         Your Web Browser             │
│                                      │
│  ┌──────────────┐  ┌──────────────┐  │
│  │   KB Chat    │  │UD Creator    │  │
│  │  & Chat UI   │  │    UI        │  │
│  └──────┬───────┘  └──────┬───────┘  │
│         │                 │          │
│  ┌──────┴─────────────────┴──────┐   │
│  │   File Summarizer UI          │   │
│  └──────────┬────────────────────┘   │
└─────────────┼────────────────────────┘
              │  Your request
              ▼
┌──────────────────────────────────────┐
│         Application Server           │
│                                      │
│  • Checks who you are (API key)      │
│  • Limits request speed              │
│  • Manages your conversation         │
│  • Remembers what was said before    │
└──────────────┬───────────────────────┘
               │
        ┌──────┴──────────────────────┐
        │                             │
        ▼                             ▼
┌──────────────────────┐   ┌──────────────────────┐
│     Your GitHub Repo │   │   GitHub Copilot AI  │
│  (finds the file/doc)│   │  (reads, thinks, replies)│
└──────────────────────┘   └──────────────────────┘
        │                             │
        └─────────────────────────────┘
                      │
                      ▼
┌──────────────────────────────────────┐
│  Answer streams back to your screen  │
│  word by word, in real time          │
└──────────────────────────────────────┘
```

In simple terms:

1. **You** type a question or paste a requirement into the web interface
2. **The server** receives it, verifies your identity, and routes it to the right AI workflow
3. **The AI** reads the relevant files from your GitHub documentation repo (and optionally Azure DevOps)

4. **The answer** flows back to your screen word by word as it's being generated — no waiting for a complete response

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Request Lifecycles — How Each Mode Works

File Summarizer

```
Browser                     Server                      GitHub API       Copilot
AI
   |                            |                            |                |
   |  Enter repo + file path    |                            |                |
   |──────────────────────────►|                            |                |
   |                            |  Verify identity           |                |
   |                            |  Validate input            |                |
   |                            |                            |                |
   |                            |  Fetch the file            |                |
   |                            |──────────────────────────►|                |
   |                            |◄──────────────────────────| raw file text  |
   |                            |                            |                |
   |                            |  Send to AI: "Summarize this"|              |
   |                            |──────────────────────────────────────────►|
   |  Summary streams word by   |                            |                |
   |  word to screen            |◄──────────────────────────────────────────|
   |◄──────────────────────────|                            |                |
   |  Done ✓                    |                            |                |
   |◄──────────────────────────|                            |                |
```

KB Chat Agent

```
Browser                     Server                  Knowledge Base        Copilot
AI
   |                            |                        |                    |
   |  Ask a question            |                        |                    |
   |──────────────────────────►|                        |                    |
   |                            |  Verify identity       |                    |
   |                            |  Look up / create      |                    |
   |                            |  your session          |                    |
   |                            |                        |                    |
   |  ⏳ "Processing..."        |                        |                    |
   |◄──────────────────────────|                        |                    |
   |                            |  Send question to AI   |                    |
   |                            |──────────────────────────────────────────►|
   |                            |                        |   AI decides which |
   |                            |                        |   KB file to read  |
   |                            |  ⚙ Fetching file...    |                    |
   |◄──────────────────────────|──────────────────────►|                    |
   |                            |                        | GitHub REST GET    |
   |                            |                        | (or cache hit)     |
```

```
   │                              │◄──────────────────────────┤  file content       │
   │                              │  File result → AI         │                     │
   │                              ├──────────────────────────────────────────────►  │
   │  ⚙ File loaded               │                           │                     │
   │◄─────────────────────────────┤                           │   AI generates      │
   │  Answer streams word by      │◄──────────────────────────────────────────────┤
   │  word to screen              │                           │                     │
   │◄─────────────────────────────┤                           │                     │
   │  Done ✓                      │                           │                     │
   │◄─────────────────────────────┤                           │                     │
```

> The AI may fetch several KB files in a row before answering. You see each fetch as a brief "⚙ loading" indicator.

UD Creator

```
Turns 1 through N  (questioning → drafting → revising)
  ↑ Same flow as KB Chat above — AI asks questions, fetches relevant KB files,
assembles draft.

Final turn — user says "Approved"
  │
  ├── AI calls create_ud_file internally
  │
  ├── Server captures the filename + full UD content
  │     → sends "UD file ready" event to browser
  │
  ├── Browser creates a download link automatically
  │     (no upload, no external service — file lives in your browser)
  │
  └── AI sends a confirmation message → stream ends ✓
```

What you see on screen during the UD workflow:

```
You:        "Build a payment retry feature. Max 3 retries, 30s intervals."

AI:         "Before I draft the UD, I have 5 questions:
             1. Scope: does this affect all payment providers or only Razorpay?
             2. Logic: what triggers a retry — network timeout only, or any
failure?
             ..."                                  ← Round 1 questions

You:        [Answers]

AI:         [Draft UD with ⚠ ASSUMED markers]       ← Full UD draft

You:        "Change section 3 — retries should stop on card-declined errors."
```

```
AI:         "Updated. Changed: Section 3 now excludes card-declined (reason code
05)."

You:        "Approved"

AI:         "☑ UD created."
            [📥 Download UD_PAY-RETRY.md]           ← Download link appears
```

SDLC Agent (Azure DevOps end-to-end)

```
Browser                     Server                    Azure DevOps         Copilot
AI
  │                         │                         │                    │
  │  Enter Work Item ID: 1234 │                       │                    │
  ├──────────────────────────►│                       │                    │
  │                         │  AI calls               │                    │
  │                         │  fetch_azure_work_item  │                    │
  │  ⚙ Fetching Work Item...  │                       │                    │
  ◄──────────────────────────┼──────────────────────►│                    │
  │                         ◄──────────────────────┤ title, description, │
  │                         │  JSON result → AI       │ acceptance criteria │
  │                         ├──────────────────────────────────────────►  │
  │  ┌──────────────────┐   │                         │   AI summarises    │
  │  │ 📋 User Story #1234 │◄──┼◄────────────────────────────────────────┤
  │  │ "Payment Retry"  │   │                         │                    │
  │  │ State: Active    │   │ "Is this correct?       │                    │
  │  └──────────────────┘   │  Shall I proceed?"      │                    │
  │                         │                         │                    │
  │  "Yes, proceed"         │                         │                    │
  ├──────────────────────────►│                       │                    │
  │                         │                         │                    │
  │        ── Phases 2-4: same as UD Creator above ──  │                    │
  │        (AI questions, KB fetches, draft, revise)  │                    │
  │                         │                         │                    │
  │  "Approved"             │                         │                    │
  ├──────────────────────────►│                       │                    │
  │                         │  AI calls               │                    │
  │                         │  create_ud_file         │                    │
  │  [📥 Download UD_WI-1234.md]│                      │                    │
  ◄──────────────────────────┤                         │                    │
  │                         │  AI calls               │                    │
  │                         │  attach_ud_to_work_item │                    │
  │  ⚙ Attaching to ADO...  │  1. POST /attachments   │                    │
  ◄──────────────────────────┼──────────────────────►│                    │
  │                         │  2. PATCH /workitems    │                    │
  │                         ├──────────────────────►│                    │
  │  ☑ UD attached to #1234 │◄──────────────────────┤                    │
  │  [View in Azure DevOps ↗]│                        │                    │
  ◄──────────────────────────┤                         │                    │
```

> The UD is both downloaded to your browser **and** permanently attached to the Azure DevOps work item — no manual upload required.

Key Concepts Explained Simply

What is a "Session"?

When you open the chat, the system creates a private "session" for you. This session remembers everything you've said in that conversation — like a phone call that stays open. You can ask follow-up questions without repeating yourself. Sessions automatically close after 30 minutes of inactivity.

What is the "Knowledge Base"?

The knowledge base is a GitHub repository containing your project's documentation — markdown files explaining how your systems work, what modules exist, API references, etc. The assistant reads an INDEX file to understand what topics live in which files, then fetches the right files when you ask a question.

What is "Streaming"?

Instead of waiting for the AI to finish its entire response and then showing it all at once, SDLC Copilot displays the answer token by token as it's generated — you see text appearing live, like watching someone type. This means you get useful information faster.

What is a "Tool"?

When the AI needs to look something up (like fetching a documentation file), it uses "tools" — predefined actions it can take. This is similar to a research assistant who knows they can look up information in the company wiki. The AI decides on its own when a tool call is needed.

What is SSE?

SSE (Server-Sent Events) is the web technology that sends the streaming text from the server to your browser in real time. It is what makes answers appear word by word.

Who Uses What Feature?

Role	Primary Feature	How It Helps
Business Analyst	SDLC Agent + UD Creator	SDLC Agent automates ADO-linked UD creation end-to-end; UD Creator handles non-ADO requirements
Developer	KB Chat + File Summarizer	Understands unfamiliar code and project components quickly
Tech Lead / Architect	KB Chat	Answers "how does X work?" questions for the team

Role	Primary Feature	How It Helps
Product Manager	KB Chat	Gets plain-English answers about system capabilities
New Team Member	KB Chat + File Summarizer	Onboards faster by exploring the codebase with AI guidance
QA Engineer	UD Creator + KB Chat	Understands requirements deeply and asks about edge cases

The UD Creation Process — Step by Step

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          UD Creator Workflow                             │
└──────────────────────────────────────────────────────────────────────────┘

   Analyst                    AI (UD Creator Mode)             Knowledge Base
      │                              │                                │
      │  Switch to "UD Creator"      │                                │
      ├──────────────────────────────►│                                │
      │                              │ Load project docs              │
      │                              ├────────────────────────────────►│
      │                              ◄────────────────────────────────┤
      │                              │                                │
      │  Paste requirement / ticket  │                                │
      ├──────────────────────────────►│                                │
      │                              │ Read relevant KB files         │
      │                              ├────────────────────────────────►│
      │                              ◄────────────────────────────────┤
      │                              │                                │
      ◄──────────────────────────────┤                                │
      │  Receives 3-7 questions      │                                │
      │  (Scope? Data? Logic? UI?)   │                                │
      │                              │                                │
      │  Answers questions           │                                │
      ├──────────────────────────────►│                                │
      │                              │  (May ask 1 more round         │
      │                              │   if still unclear)            │
      ◄──────────────────────────────┤                                │
      │  Receives complete UD draft  │                                │
      │  (⚠ marks uncertain items)   │                                │
      │                              │                                │
      │  Reviews and gives feedback  │                                │
      ├──────────────────────────────►│                                │
      ◄──────────────────────────────┤                                │
      │  Revised UD with change notes │                               │
      │                              │                                │
      │  "Approved" / "Looks good"   │                                │
      ├──────────────────────────────►│                                │
```

```
        │                                │ Generates UD_TICKET-123.md    │
        │◄───────────────────────────────┤                               │
        │  Download link appears         │                               │
        │  in chat                       │                               │
```

What the AI will ask about (the 7 question categories):

Category	What the AI is asking	Example
Scope	What's in, what's out	"What exactly should this NOT affect?"
Data / Mapping	Where data comes from and goes	"Which system is the source? What field maps to what?"
Business Logic	Rules, conditions, edge cases	"What happens if the user has no account yet?"
Integration	External systems involved	"Which API does this call? How often?"
UI / UX	Screens, forms, flows	"Is this a new screen or changing an existing one?"
Testing	How to know it's working	"What does a successful test look like?"
Dependencies	What must exist first	"Does the database table already exist?"

What Makes It Reliable?

Your Conversation Is Private

Each chat session is locked to your account. Other users cannot see or access your conversation even if they somehow knew your session ID.

The AI Stays on Topic

The KB Chat and UD Creator modes only answer based on your actual project documentation. If the information isn't in your knowledge base, the AI says so — it does not make things up or guess.

It Handles Slow Networks

The server sends regular "heartbeat" signals to keep streaming connections alive even through corporate proxies that would otherwise cut off slow connections.

It Protects Against Overuse

Rate limits prevent any single user from overwhelming the system with too many requests. The server also caps the total number of active conversations.

Setting It Up — What You Need

To run SDLC Copilot, your team needs:

1. **A GitHub account** with an active **GitHub Copilot subscription**

2. **A GitHub Personal Access Token** (PAT) — a secure credential that lets the app read your repositories
3. **A documentation repository** on GitHub containing your project's markdown files (for KB Chat / UD Creator)
4. **An** `INDEX.md` in that repo — a table of contents that maps topics to file paths. This helps the AI navigate your docs efficiently.
5. **Node.js 18+** installed on the server where you'll run the application

For **SDLC Agent mode** (Azure DevOps integration), you additionally need:

6. **An Azure DevOps Personal Access Token** with **Work Items (Read, Write)** scope
7. Your **Azure DevOps organization name** and **default project name**

Frequently Asked Questions

Q: Does the AI read my code, or just documentation? The File Summarizer can read any file (code, config, markdown). The KB Chat and UD Creator only read files from your designated documentation repository, and only files with allowed extensions (markdown by default).

Q: Can it access private repositories? Yes. You provide a GitHub Personal Access Token with `repo` scope, and the app uses it to authenticate with GitHub's API directly.

Q: Is my conversation stored anywhere? Conversations are held in the server's memory for the duration of your session (up to 30 minutes of inactivity). They are not written to a database or file. Restarting the server clears all sessions.

Q: What happens if the AI doesn't know the answer? For KB Chat and UD Creator, the AI will explicitly say the information is not available in the knowledge base rather than guessing. An INDEX.md file in your KB repo helps it locate obscure topics before giving up.

Q: Can multiple people use it at the same time? Yes. Each user gets their own independent session. The system supports up to 500 concurrent sessions by default (configurable).

Q: Does the SDLC Agent need Azure DevOps? Yes. The SDLC Agent mode requires an Azure DevOps Personal Access Token (`ADO_PAT`) along with the organization name (`ADO_ORG`) and project name (`ADO_PROJECT`) configured in the server's environment. If these are not set, the server starts normally but SDLC Agent tool calls will return an error. KB Chat and UD Creator are unaffected.

Q: How do I update the knowledge base? Commit updated markdown files to your KB GitHub repository. The next new chat session will pick up the latest versions of core files. Files fetched on-demand from the knowledge base are cached for performance but the cache is cleared when the server restarts.

Q: What is an Understanding Document (UD)? A UD is a structured document that captures a software requirement in enough detail for developers to implement it without ambiguity. It typically includes scope, business rules, data mappings, UI changes, integration points, and testing criteria. The UD Creator automates the process of gathering this information through targeted questions.

Roadmap — What's Coming Next

> **Status key:** ☑ Shipped · 🔨 In Progress · 📋 Planned

```
TODAY                PHASE 0              PHASE 1              PHASE 2
PHASE 3               PHASE 4
──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
File Summarizer ☑    Agent Loop 📋        Structured Logging 📋  Redis Sessions
📋     5 New AI Modes 📋     Error Recovery 📋
KB Chat ☑           Planning Tool 📋     PostgreSQL DB 📋       SSE EventBus 📋
PR Review 📋          Full Security 📋
UD Creator ☑        Reflection Tool 📋   Message History 📋     Artifact Store
📋     Test Generator 📋     Conversation Sidebar 📋
SDLC Agent ☑        Provider Layer 📋    Multi-Project KB 📋    Horizontal
Scale 📋   HLD Generator 📋      Markdown Editor 📋
                    Skills System 📋     Context Survival 📋
Release Notes 📋      PDF / Confluence Export 📋
                    Security Fixes 📋
OAuth / SSO 📋        Per-tenant Limits 📋
                    Workspace Config 📋
Role-Based Access 📋
```

Phase 0 — Agent Architecture Foundation

Prerequisite for all future phases. Makes SDLC Copilot a true autonomous agent, not just a chatbot.

Improvement	What it means for you
Agent Loop	AI can plan multi-step tasks, reflect on its progress, and self-correct without you nudging it
Planning & Reflection Tools	AI writes an explicit plan before executing it — you can see and validate its intended steps
Provider Abstraction	Swap or mix AI models without any code change
Skills System	New capabilities are drop-in plugins — adding PR Review or Test Gen doesn't touch existing code
Workspace Config	One deployment serves multiple teams with different KB repos and tool sets
Security Fixes	Prompt injection defence, API key moved server-side, dev-mode auth lock

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Phase 1 — Production Foundation

Makes the system observable, persistent, and multi-project ready.

Improvement	What it means for you
Structured Logging	Every AI decision, tool call, and error is logged with full context for debugging

Improvement	What it means for you
PostgreSQL + Message History	Conversations survive server restarts; continue where you left off
Multi-Project KB	One deployment, many teams — each project has its own knowledge base and settings
Artifact Storage	Generated UDs are saved server-side; no more lost files on page refresh

Phase 2 — Resilience & Scale

Prepares for production load and multi-instance deployments.

Improvement	What it means for you
Redis Session Metadata	Multiple server instances with sticky routing — horizontal scaling without data loss
SSE EventBus	Streaming events are decoupled from the HTTP layer — logs, metrics, and UI all get the same events reliably
Artifact Store	Download any past UD by ID, not just the most recent one in your browser

Phase 3 — SDLC Expansion

Extends AI coverage from requirements all the way through release.

```
Requirements  →  Design  →  Development  →  Testing  →  Release  →  Maintenance
──────────────────────────────────────────────────────────────────────────────
KB Chat ☑          HLD 📋       PR Review 📋    Test Gen 📋   Release    Incident
UD Creator ☑                   File Sum ☑                  Notes 📋    Analysis
📋
SDLC Agent ☑
```

New Mode	What it does
PR Review	Analyses a GitHub pull request against your KB coding standards, flags gaps
Test Generator	Creates test cases from a UD or requirement description
HLD Generator	Produces a High-Level Design document from requirements
Release Notes	Summarises all commits and PRs between two release tags
Incident Analyser	Diagnoses root cause from an incident description using KB context
OAuth / SSO + RBAC	Login with your company SSO; role-based access (Viewer / Analyst / Developer / Admin)

Phase 4 — Polish

Last-mile quality, security hardening, and power-user UI features.

Improvement	What it means for you
Circuit Breaker	GitHub / ADO outages degrade gracefully — cached data serves instead of errors
Full Prompt Injection Defence	All 5 security layers active: input sanitisation, role-lock prompts, context isolation, output validation, tool guardrails
Conversation Sidebar	Browse and restore past sessions by project from the left panel
Markdown UD Editor	Edit your UD inline in the chat before approving, without copy-pasting to another tool
Export Options	Download as PDF, push directly to Confluence, or create a Jira ticket
Agent Progress Indicator	See each planning step as it executes — know exactly what the AI is doing and why

Glossary

Term	Plain English
API	A way for software programs to talk to each other
GitHub	A platform for storing and sharing code online
Azure DevOps	Microsoft's platform for project management and work item tracking
Copilot SDK	A software toolkit that connects to GitHub's AI
SSE	Technology that streams text to your browser in real time
LLM	Large Language Model — the AI that reads and generates text
Session	A private, remembered conversation between you and the AI
Knowledge Base (KB)	Your GitHub repo containing project documentation
Tool call	An action the AI takes automatically, like fetching a file
UD	Understanding Document — a structured software requirement specification
Work Item	A ticket in Azure DevOps (User Story, Bug, Task, Feature, etc.)
Rate limit	A cap on how many requests can be made in a given time window
PAT	Personal Access Token — a secure credential (GitHub or Azure DevOps)
Monorepo	A single code repository containing multiple related projects